

AVL TREE SIMULATION AND TRAVERSAL

Tool-VisuAlgo

CO3-AT2

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OPERATIONS:

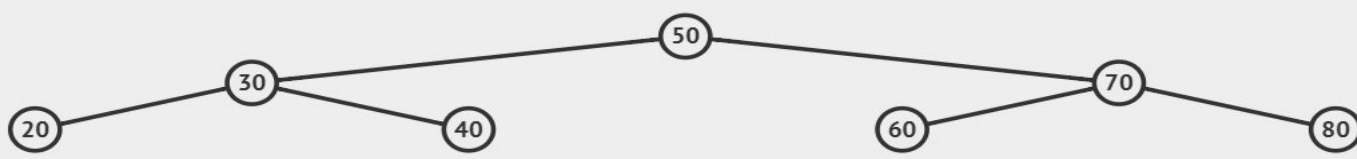
1. INSERTION
 2. DELETION
 3. SEARCHING
 4. TRAVERSING
 - i. IN-ORDER
 - ii. PRE-ORDER
 - iii. POST-ORDER
 5. ROTATION
 - i. LEFT-SIDE
 - ii. RIGHT-SIDE
- 

1.INSERTION:

- Select Insert in the AVL simulator and enter 50, 30, 70, 20, 40, 60, 80 one by one.
- The simulator places each value according to the BST property while maintaining AVL balance.
- The final tree contains 3 levels with 50 as the root.
- No rotation is required because the tree remains balanced.

VISUALGO.NET / en / bst BINARY SEARCH TREE AVL TREE Exploration Mode [LOGIN](#)

N=7, h=2



```
graph TD; 50((50)) --- 30((30)); 50((50)) --- 70((70)); 30((30)) --- 20((20)); 30((30)) --- 40((40)); 70((70)) --- 60((60)); 70((70)) --- 80((80));
```

Insert(80)

The tree is balanced.

v = 80 [Go](#)

- [Toggle BST Layout](#)
- [Create](#)
- [Search\(v\)](#)
- [Insert\(v\)](#)
- [Remove\(v\)](#)
- [Predec-/Succ-essor\(v\)](#)
- [Select\(k\)](#)
- [Traverse\(root\)](#)

1.0x 1x

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2. DELETION:

- Select Delete and enter 20.
- The simulator follows $50 \rightarrow 30 \rightarrow 20$ and removes node 20.
- Since 20 is a leaf node, it can be directly deleted.
- The remaining AVL tree stays balanced.

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N=6, h=2

```
graph TD; 50((50)) --- 30((30)); 50 --- 70((70)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

Remove(20)

The tree is balanced.

v = 20 Go

- Toggle BST Layout
- Create
- Search(v)
- Insert(v)
- Remove(v)
- Predec-/Succ-essor(v)
- Select(k)
- Traverse(root)

1.0x 1x

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3. SEARCHING:

- Select Search and enter 60.
- The simulator compares 60 with 50 and moves right to 70.
- Since $60 < 70$, it moves left and reaches 60.
- Search path: $50 \rightarrow 70 \rightarrow 60$ — 60 found.

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N=7, h=2

```
graph TD; 50((50)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80)); style 50 stroke:#f96; style 70 stroke:#f96; style 60 stroke:#f96;
```

Search(60)

Value 60 is found.

v = 60 Extreme:

4. TRAVERSING:

i. IN-ORDER: (Left – Root – Right)

Visualgo.net interface showing a Binary Search Tree (BST) visualization. The tree is a balanced AVL tree with root 50, left child 30, right child 70, and leaf nodes 20, 40, 60, and 80. The tree is labeled "N=7, h=2".

Operations menu (left sidebar):

- Tog
- Cre
- Se
- Ins
- Ret
- Pre
- Se
- Tr

Operation performed: **Insert(20)**

Message: **The tree is balanced.**

Navigation controls: 1.0x, play/pause, stop, next, previous, progress bar.

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ii. PRE-ORDER: (Root – Left – Right)

Visualgo.net interface showing a Binary Search Tree (BST) visualization.

Browser tabs: Binary Search Tree, AVL Tree - V, AVL Tree vs Binary Tree

URL: visualgo.net/en/bst?utm_source=chatgpt.com

Navigation: VISUALGO.NET / en / bst, BINARY SEARCH TREE, AVL TREE

Mode: Exploration Mode, LOGIN

Tree Structure (N=7, h=2):

```
graph TD
    50((50 root)) --- 30((30))
    50 --- 70((70))
    30 --- 20((20))
    30 --- 40((40))
    70 --- 60((60))
    70 --- 80((80))
```

Inorder Traversal

Inorder traversal of the whole BST is complete.
Visitation sequence: 20,30,40,50,60,70,80.

```
if this is null
    return
Inorder(left)
visit this
Inorder(right)
```

1.0x 1x

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iii. POST-ORDER: (Left – Right – Root)

Binary Search Tree, AVL Tree - V

Ask Gemini

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BINARY SEARCH TREE AVL TREE

Exploration Mode

LOGIN

N=7, h=2

```
graph TD; 50((50 root)) --- 30((30)); 50 --- 70((70)); 30 --- 20((20)); 30 --- 40((40)); 70 --- 60((60)); 70 --- 80((80));
```

<

Toggle BST Layout

Create

Search(v)

Insert(v)

Remove(v)

Predec-/Succ-essor(v)

Select

Traverse(root)

Preorder Traversal

Preorder traversal of the whole BST is complete.
Visitation sequence: 50,30,20,40,70,60,80.

```
if this is null
  return
visit this
Preorder(left)
Preorder(right)
```

>

5. ROTATION:

i. LEFT-SIDE:

Binary Search Tree, AVL Tree - V

AVL Tree vs Binary Tree

Ask Gemini

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=0, h=0 (empty BST)

Remove(30)

The tree is balanced.

```
remove v
check balance factor of this and its children
case1: this.rotateRight
case2: this.left.rotateLeft, this.rotateRight
case3: this.rotateLeft
case4: this.right.rotateRight, this.rotateLeft
this is balanced
```

1.0x

1x

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ii. RIGHT-SIDE:

Binary Search Tree, AVL Tree - V

AVL Tree vs Binary Tree

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=0, h=0 (empty BST)

Remove(30)

The tree is balanced.

1.0x

1x

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Thank
you